

The empirical recurrence for OEIS A304146, recovered from its tail

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Abstract

OEIS A304146 records “Empirical recurrence of order 93” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 150$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 97$. Its last coefficient is $c_{93} = 9216 \neq 0$, so no further tail satisfies anything shorter; any order-93 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A304146 is “Number of $n \times 5$ 0..1 arrays with every element unequal to 1, 2, 3 or 6 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

3, 70, 113, 157, 1198, 3484, 5336, 24401, ...

The entry states:

Empirical recurrence of order 93 (see link above)

As of the “Last modified” line on the live entry (May 07 2018 08:53:55, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 150$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 150$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 93: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 93.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 300$ exact terms from $n = 4$ on, it returns order 93 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{93} c_i a(n-i),$$

c_1	2	c_{25}	25207240	c_{49}	−15958691969	c_{73}	−883687900
c_2	7	c_{26}	−221602597	c_{50}	9093625564	c_{74}	−394088096
c_3	46	c_{27}	−440130523	c_{51}	30453906891	c_{75}	9615000
c_4	−61	c_{28}	−178245317	c_{52}	22162376283	c_{76}	97694708
c_5	−458	c_{29}	789208422	c_{53}	−4143979189	c_{77}	16434192
c_6	−1233	c_{30}	1408304521	c_{54}	−26325957010	c_{78}	19108744
c_7	1150	c_{31}	707921254	c_{55}	−22365734225	c_{79}	46852416
c_8	10757	c_{32}	−1607031718	c_{56}	−1069755320	c_{80}	43275480
c_9	23412	c_{33}	−3089658806	c_{57}	18289152749	c_{81}	650736
c_{10}	−13073	c_{34}	−1984942421	c_{58}	18887985226	c_{82}	−25826432
c_{11}	−139629	c_{35}	1681644054	c_{59}	4218020178	c_{83}	−19938784
c_{12}	−266459	c_{36}	4728115955	c_{60}	−11762068252	c_{84}	−3560704
c_{13}	71133	c_{37}	3964154734	c_{61}	−14634684762	c_{85}	5804576
c_{14}	1060892	c_{38}	428579806	c_{62}	−4895806429	c_{86}	4214240
c_{15}	1745678	c_{39}	−4128677029	c_{63}	7486300578	c_{87}	1206208
c_{16}	−81383	c_{40}	−5044108578	c_{64}	10450298908	c_{88}	−645056
c_{17}	−4638855	c_{41}	−4825439590	c_{65}	4076926169	c_{89}	−424832
c_{18}	−6041124	c_{42}	−1679981005	c_{66}	−4201092660	c_{90}	−173056
c_{19}	−700361	c_{43}	2168101164	c_{67}	−6339623704	c_{91}	27648
c_{20}	8806987	c_{44}	9340077042	c_{68}	−2684838159	c_{92}	13824
c_{21}	3142372	c_{45}	13234010070	c_{69}	1725707064	c_{93}	9216
c_{22}	650854	c_{46}	5973635946	c_{70}	2926236280		
c_{23}	22536224	c_{47}	−11170890362	c_{71}	1314138192		
c_{24}	76705820	c_{48}	−25392314800	c_{72}	−407805310		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 97$, and it is the recurrence OEIS A304146 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{93} - c_1x^{93-1} - \dots - c_{93}$. Its constant term is $-c_{93} = -9216$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 93 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 93, so $\deg q = 93$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 93, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 25 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 93, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 9216.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-93 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A304146.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.